

W4

BLOCK 1 · CLASS 4 · PRACTICAL

Rules, judge, loop.

Last week you watched a feature get built. Today you build one.

Today.

- 1 Check-in
- 2 What an agent actually is
- 3 Three standards
- 4 Git: one change, start to finish
- 5 Build one, live — then you build one
- 6 Retro: one new rule

Your GitHub page.

- A branch for feature 1.
- A merged pull request.

Not there yet? Start there today.

Chat model, or agent.

A CHAT MODEL

Text in, text out.

It has never seen your folder.

AN AGENT

The same model, with hands.

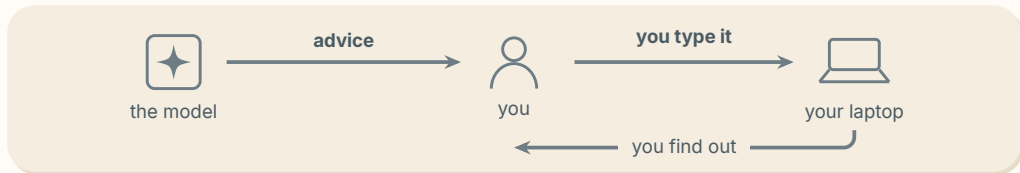
Reads. Edits. Runs. Sees the result.
Chooses again.

Agent = model + tools + a loop.

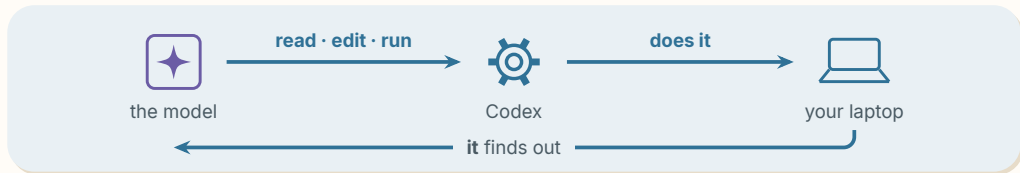
WHAT IS AN AGENT?

Tools.

WITHOUT TOOLS

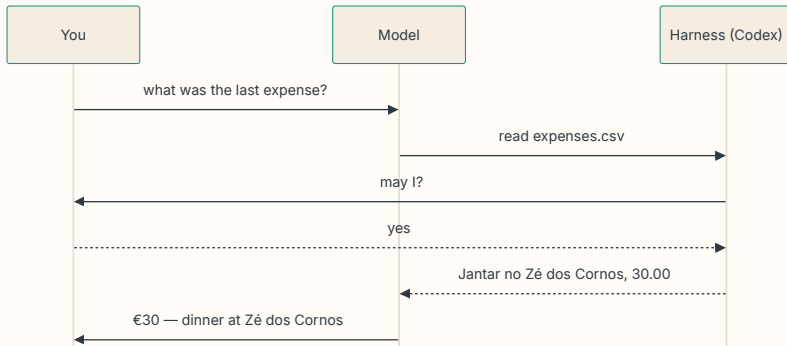


WITH TOOLS



The same model both times. What changes is who finds out that it was wrong.

A tool call.



Three tools cover nearly everything it does.

One exchange, in full.

YOU TYPE what was the last expense?

THE MODEL EMITS `{"tool": "read_file", "path": "data/expenses.csv"}`

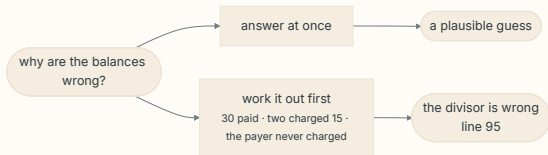
CODEX ASKS *Codex wants to read* `data/expenses.csv` *. Allow?*

CODEX PASTES BACK `29,3,8,Jantar no Zé dos Cornos,30.00,2026-08-25`

THE MODEL ANSWERS €30 — dinner at Zé dos Cornos, on 25 August.

It never saw the file. It asked, you said yes, and the line came back.

Reasoning.



WHAT IT IS

Work it out in steps, *then* decide — as you would on paper.

THE PROBLEM IT SOLVES

A model answers in one pass — the same effort for $2 + 2$ as for your bug. The steps it writes **are** the extra thinking.

WHY IT MATTERS

A guess at step one is still wrong at step ten.

WHAT IS AN AGENT?

Reasoning, watched.

YOU

Make it so Ana doesn't owe anything.

THINKING

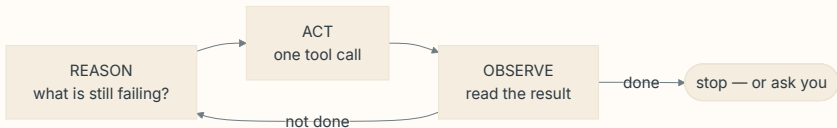
That could mean delete her expenses, mark her settled up, or take her out of the split. Two of those change what everyone else owes. One doesn't.

CODEx

Marked Ana settled up, so the other balances are untouched. Deleting her expenses would move theirs too — tell me if you meant that.

WHAT IS AN AGENT?

The ReAct loop.



THE HARNESS

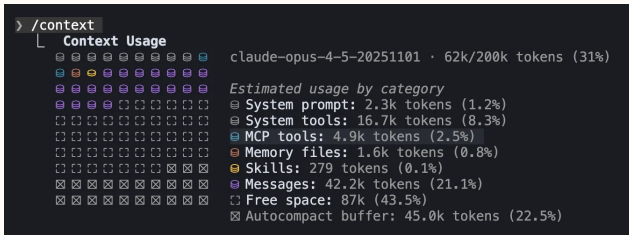
It runs the tool, asks your permission and feeds the result back. Codex is one; so are Claude Code and Cursor.

WHY IT MATTERS

They all run this same loop. Learn it once, and the tools are interchangeable.

Yao, Zhao, Yu, Du, Shafran, Narasimhan & Cao (2022). *ReAct: Synergizing Reasoning and Acting in Language Models*. ICLR 2023 · arxiv.org/abs/2210.03629

Context.



WHAT IT IS

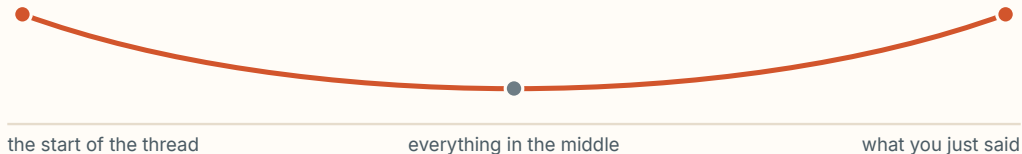
Everything the model can see at once: your messages, every file it read, every command's output. The window has a fixed size.

WHY IT MATTERS

As it fills, the earliest instructions get crowded out. A long thread does not get wiser. It gets vaguer.

Context rot.

HOW WELL IT USES WHAT IS THERE



IT IS MEASURED

Eighteen models tested. Every one got worse as the input grew, well short of the limit.

LOST IN THE MIDDLE

It uses the start and the end. The middle goes soft — and that is where your bug is.

SO: COMPACT

Keep a summary and what just happened, drop the rest. A million-word window is not a memory.

Liu, N. F. et al. (2024). *Lost in the Middle: How Language Models Use Long Contexts*. TACL 12 · arxiv.org/abs/2307.03172 · Chroma (2025), *Context Rot*

The long run-up.



The ideas were all there. **Something else was missing.**

Why it happened now.

COMPUTE

Arithmetic got cheap

In 2007 graphics cards became programmable. In 2012 a network trained on two of them won the image contest.

DATA

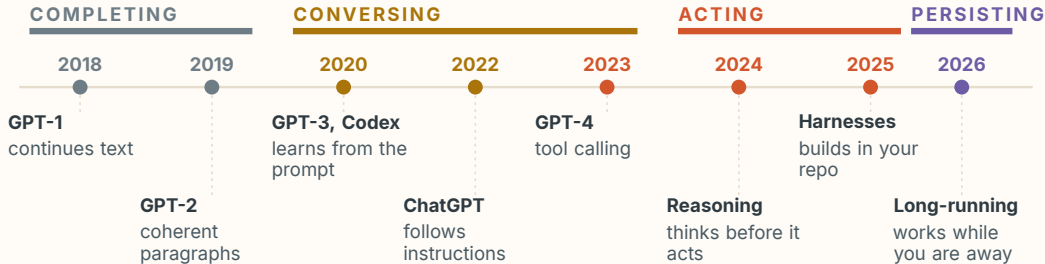
The web became a corpus

Since 2007 a non-profit has crawled the public web and given it away. Billions of pages, free.

Neural nets are not new. **Cheap parallel arithmetic and a scraped web are.**

NVIDIA CUDA (2007) · Krizhevsky, Sutskever & Hinton, AlexNet (2012) · Common Crawl (founded 2007, public crawls from 2011)

2018 onwards.



The job changed twice: when it learned to take **instructions**, and when it got **tools and a loop**.

3

THE CORE OF THE CLASS

Three standards.

A rulebook it reads. A judge it cannot argue with. A loop that knows when to stop.

AGENTS.md

WHAT IT IS

A README for agents.

Every session starts by reading it.

WHAT GOES IN

What it cannot guess.

Commands. Boundaries.
How to report back.

WHAT STAYS OUT

What it can read.

Your code, explained.
Every extra line hides a real one.

It lives in git. When the agent misbehaves, the fix is usually a line here.

Three rules.

BOUNDARY

One branch per spec.

Never straight to `main`.

EVIDENCE

Show me the test output.

The output itself, not a summary of it.

OFF-LIMITS

Never touch the tests.

If one looks wrong, say so and stop.

Exact wording on the Week 4 page. Add them, read the diff, commit, push.

The judge.

THE PROBLEM

So give it a check.

Failing tests are the to-do list. Green is the finish line.

THE RULE

Editing a test is cheating.

It is not finishing.

EVIDENCE

Show me the output.

Anyone can type "all tests pass".

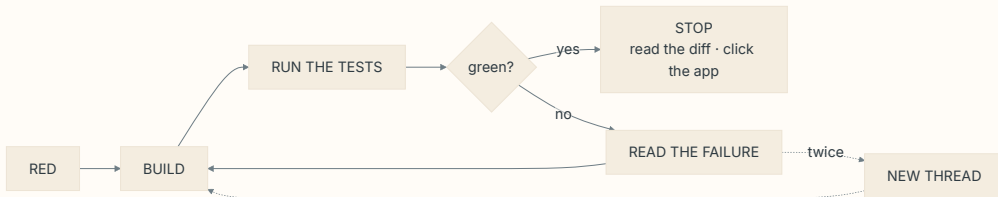
AFTER GREEN

Read the diff. Click the app.

App wrong, tests green? You found a missing test.

Decide what *done* means before you start.

The loop.



ONE TASK PER LOOP

One bug, or one spec.

LET IT LOOP

It reads its own failures.
You read the result.

KNOW WHEN TO STOP

Green. Or twice round
on the same failure, and
then a new thread.

The shape of an ask.

1

The task

One thing.
Name the spec.

2

The boundary

This branch.
Never the tests.

3

The judge

Run the tests
until they pass.

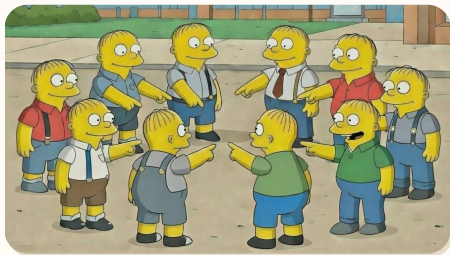
4

The evidence

The output, and
every file
touched.

Written out in full on the Week 4 page. Copy it; change two names.

Ralph.



WHAT IT IS

The same prompt, handed to a fresh agent again and again, until the judge says green.

WHY A CROWD

Every run is a new one, and none remembers the last. What they need to know has to be written down.

Nobody is watching.

```
while ;; do cat PROMPT.md | claude-code ; done
```

Published in 2025. That is the entire technique.

THE PROMPT

One file, read again from scratch every time. It is the only instruction anyone gives.

THE REPO

Nothing survives a run except what was written to disk. The rulebook and the tests are the memory.

THE CATCH

Nobody reads anything until it stops. A weak judge buys you hours of confident wrong.

/goal

```
/goal every test in tests/ passes, and nothing in tests/ changed
```

2026. The shell loop is now one sentence, in the tool.

WHO DECIDES

After each turn a *second*, smaller model reads the conversation and rules: not yet, done, or impossible. Not the one doing the work.

WHAT IT CAN SEE

It cannot run your tests. It only reads what the agent put on screen. So the finish line has to be something the agent can show you.

STILL YOURS

The tool runs the loop now. Nobody has automated deciding what *done* means.



HOW A TEAM STAYS SANE

One change, start to finish.

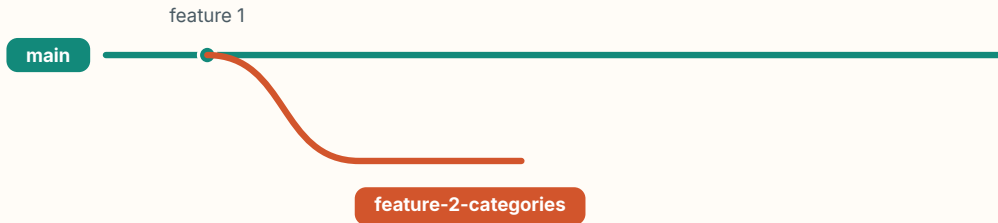
`main` is the version everyone trusts. Here is how a change gets into it.

main



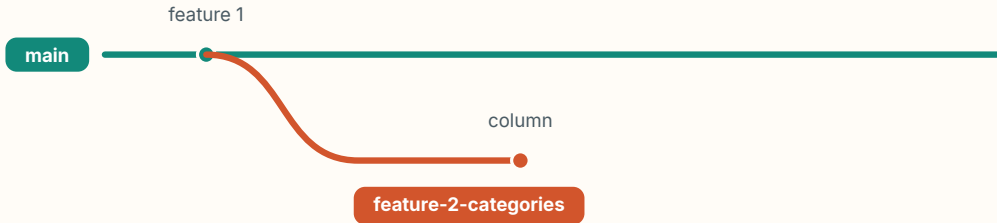
One version, one history, one truth. No folder full of *model_v3_FINAL_jm*.

A branch.



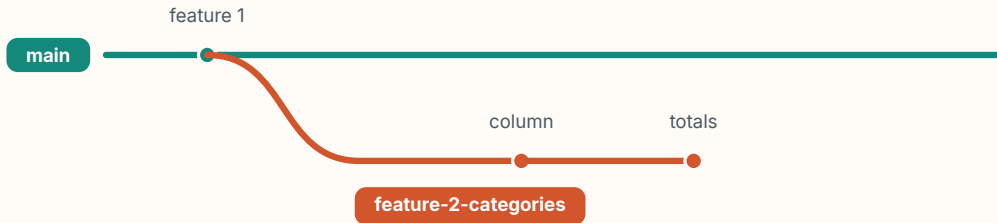
A **branch** is your own copy to work in, named after the job. Everyone else carries on from `main`, undisturbed.

A commit.



A **commit** is a checkpoint with a note saying what changed. Small ones. You can go back to any of them.

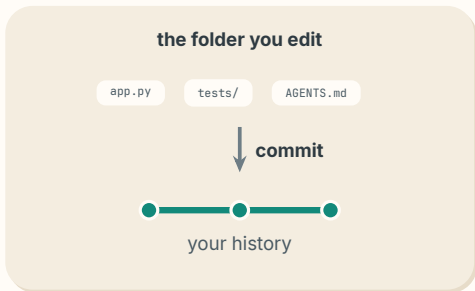
Working on a branch.



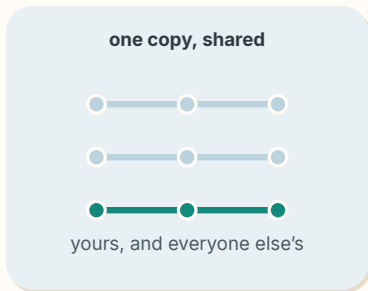
Three commits in and half of it broken. Everyone else's copy is untouched.

Local, and remote.

YOUR LAPTOP · LOCAL



GITHUB · THE REMOTE

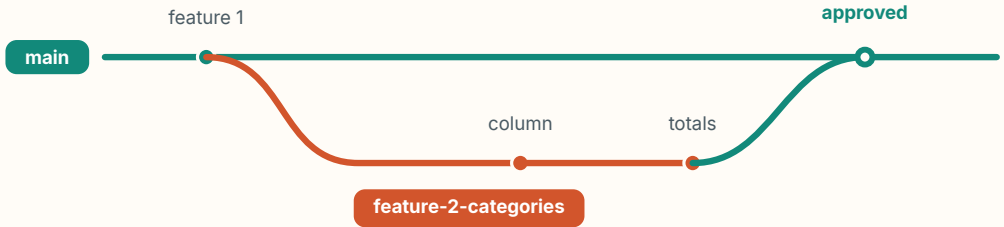


push
→
your branch, up

pull
←
their work, down

Push sends your branch to the copy everyone shares, where a **pull request** asks someone to read the diff first. Today that someone is you, reviewing the agent.

Merge.



Merge is the sign-off. `main` now has your work, and the next person starts from there.

Feature 2, live.

- 1 **Branch.** Named after the spec.
- 2 **Red first.** See what is missing before anything is written.
- 3 **New thread. Paste the ask.** Then sit on your hands.
- 4 **Watch it loop.**
- 5 **Green — now don't trust it.** Read the diff. Click the app.
- 6 **Commit, push, pull request, merge.**

One bug, then feature 2.

SPLITIT

Bug 2, then feature 2

Removing a member crashes the page.
Then expense categories.

TINY CRM

Bug 3, then feature 2

"Won this month" is always zero. Then
conversion by source.

Files, test names and the ask: Week 4 page.

When it goes wrong.

- 1 **It edited a test.** Discard. Say the rule. Run again.
- 2 **"Done" — tests red.** Ask for the output.
- 3 **Same failure, three times.** Stop. Read it yourself. New thread.
- 4 **It fixed three other things.** Discard. One task per loop.
- 5 **Green, app wrong.** Write down what the judge missed.

Every line here is a candidate rule for your `AGENTS.md`.

Write one rule.

- The worst thing it did today — one sentence.
- Turn it into a rule you could check.
- Commit it to `AGENTS.md`, with the reason.

A rulebook grows one line at a time, from things that went wrong.

Before next week.

- Feature 2: green, merged, pushed.
- One new rule in `AGENTS.md`.
- Push before you close the laptop.

Rulebook · judge · loop.